

Department of Leather Engineering
Khulna University of Engineering & Technology
B. Sc. Engineering 4th Year 2nd Term Examination-2020
Production & Quality Control
LE 4217

Time: 1 hour and 30 minutes.

Full Marks: 120

- N.B. i) Answer any **TWO** questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable data if any missing.

SECTION-A

- 1(a) ABC footwear company has projected sales of 290, 520, 390, 110, 220 and 290 pair of shoes for the six months January to June and the number of working days in the corresponding months are 22, 19, 21, 21, 22, and 20, respectively. The company at present employs 20 workers and each pair requires 10 labor hours for its production. The hiring costs are Tk. 600 and layoff costs are Tk. 700 per person. Company policy is to retain a safety stock equal to 20% of the monthly forecast. There are currently 40 pair in stock carried at Tk. 4 per pair-month. A cost of Tk. 20 and Tk. 70 per pair of shoes have been assigned for stock out and out-sourcing respectively. Straight time (8 hours) cost is Tk. 12/hour. Suggest an effective aggregate plan based on the following strategies: (i) Vary the workforce size to accommodate the demand, (ii) Constant workforce of 20 and build inventory or incur stock out cost, and (iii) Minimum constant workforce. 24
- 1(b) How routing, scheduling and loading are inter-connected? 06
- 2(a) Suppose you are a footwear expert willing to establish a company near Khulna as an entrepreneur with an initial production target of 1800 pair of shoes per month with six varieties. (i) Discuss about the production elements that you should prepare first, (ii) Which production system and layout you will choose for this production? Defend your comments. 14
- 2(b) State the law of variable proportions. Describe the causes of variation among the stages of law of return to sale. 11
- 2(c) Outline the material properties that can affect the type of material handling transport and storage equipment. 05
- 3(a) What is method study? Write down the procedures involved in work measurement. 12
- 3(b) How both art and science involved in predicting future events? 03
- 3(c) Given below are the sales pattern of a reputed company, manufacturing leather wallet. Trends of amount spent on advertising is also depicted. Find out the correlation between sales of wallet and advertising expenditure. Also compute the estimates of sales when expenditure for the marketing purpose will rise to 60 thousand taka. 15

Month	Sales (Thousand Tk.)	Advertising (Thousand Tk.)
January	1063	34
February	1190	43
March	1298	48
April	1421	52
May	1440	57
June	1518	58

SECTION-B

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| 4(a) | Write down the basic difference between quality assurance and quality control. | 05 |
| 4(b) | What is Total Quality Management (TQM)? As a leather engineer, how will you implement TQM in a footwear industry? | 10 |
| 4(c) | What do you mean by Six Sigma? Write down the methods by which you can improve six sigma projects. | 10 |
| 4(d) | What are the main reasons why TQM fails in an industry? | 05 |
| 5(a) | Describe the relation between an industry and environment. | 06 |
| 5(b) | Write down some health and safety measures followed in lasting and moulding department for footwear industry. | 08 |
| 5(c) | How you can apply quality control in different department of leather products industry? | 06 |
| 5(d) | ABC footwear company estimates its carrying cost at 20% and ordering cost at \$10 per order. The estimated annual requirement is 50,000 units at a price of \$5 per unit. Determine:
(i) What is Economic Order Quantity (EOQ)? (ii) Number of orders to be placed in a year? | 10 |
| 6(a) | Why inventory management is important? Describe the ABC inventory control techniques. | 12 |
| 6(b) | Define Gantt Chart. Mention some benefits of Gantt Chart. | 08 |
| 6(c) | Why Zero defects concept is necessary to maintain quality in an industry? Describe it in respect of goals and principles of zero defects. | 10 |